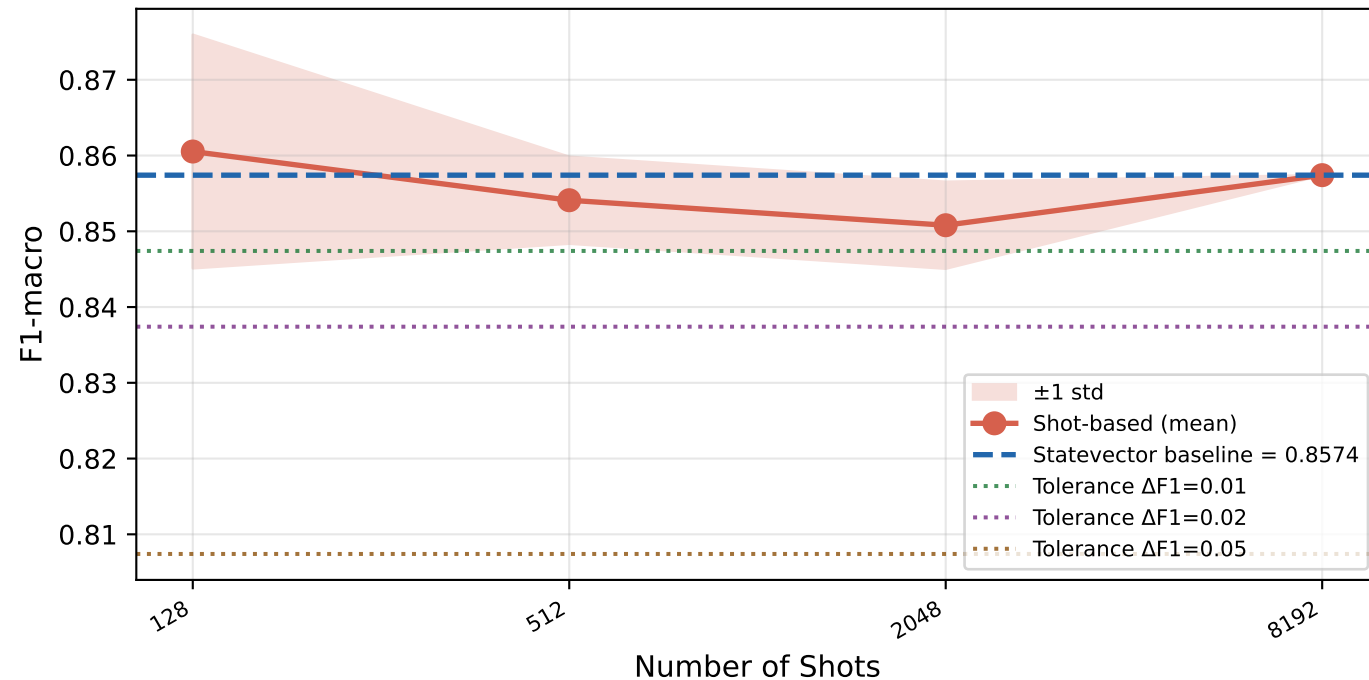


C2.5 — Shot Convergence Analysis

Minimum shots for $\Delta F1 < \text{tolerance}$ (NISQ feasibility bound)

(a) Convergence of F1-macro to Statevector Baseline



(b) NISQ Feasibility Threshold Table

Tolerance	Min Shots	Actual $\Delta F1$	Fro. Sim
$\Delta F1 < 0.01$	128	-0.0031	0.9974
$\Delta F1 < 0.02$	128	-0.0031	0.9974
$\Delta F1 < 0.05$	128	-0.0031	0.9974

NISQ Feasibility Summary

- SV baseline F1 = 0.8574
- Max F1 degradation = 0.0066 (shots=128)
- Min F1 degradation = -0.0031 (shots=8192)

- Kernel MAE (shots=128): 0.01539
- Kernel MAE (shots=8192): 0.00186

Shot noise follows $\sim 1/\sqrt{\text{shots}}$ theory.
See threshold table above for recommended deployment shots.